

Energy-Aware Queueing and Bandwidth Allocation for On-Orbit Federated Learning

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Abstract

The growth of AI inference is placing unprecedented demand on terrestrial data centers, prompting interest in on-orbit compute aboard nanosatellite constellations. Industry initiatives from Starcloud, Google Project Suncatcher, and SpaceX address hardware feasibility and constellation scale, while recent orbital federated-learning studies adapt training protocols to sparse ground contact. Neither line of work characterizes per-satellite gradient-buffer dynamics under coupled energy and link constraints. We study the scheduling problem faced by a single nanosatellite engaged in federated learning in low Earth orbit, where gradient updates must be transmitted over intermittent laser inter-satellite links subject to a hard energy budget. We model battery, illumination, and contact states as a discrete-time Markov chain and derive the effective server uptime P_{ready} in closed form. The gradient buffer forms a queue served by an on-off process driven by this chain. We solve the joint Markov-modulated queue exactly for mean update staleness; a single-geometric vacation approximation underestimates staleness by roughly a factor of two because off-periods are over-dispersed across contact-limited and battery-limited phases. The exact solution agrees with a discrete-event simulator across the full stable load range (38.9 slots analytically versus 39.1 ± 2.9 simulated at the nominal load). At the constellation level, per-satellite backlogs factor under independent eclipses and decouple when satellites in the same plane share illumination transitions. We formulate bandwidth allocation as a constrained Markov decision process over 1,764 joint states and solve it by Lagrangian relaxation and value iteration. The resulting policy maintains the battery above its energy floor in every simulation seed while delivering 14.7% higher gradient throughput than the best fixed heuristic.

1 Introduction

The rise of artificial intelligence, and of large language models in particular, is pushing terrestrial computing infrastructure toward physical and thermodynamic limits. In 2024, a single query to a model such as ChatGPT consumed an estimated 2.9 watt-hours of electricity, roughly ten times the energy of a standard web search [1]. As model size and query volume grow in parallel, this per-transaction footprint compounds faster than many regional grids can accommodate. United States data center electricity demand is projected to increase by 130% by 2030 [2, 3]. Power is only one constraint. Terrestrial facilities also compete for land, water, and cooling capacity at a scale that is increasingly difficult to sustain.

A modern AI-focused data center can consume up to 5 million gallons of water daily for cooling, which is equivalent to the usage of a city of 50,000 people [4]. At the same time, it requires the power infrastructure of 100,000 homes. Furthermore, these facilities consume hundreds of acres of land, permanently displacing agricultural and natural environments [5]. Attempts to mitigate these constraints by relocating infrastructure to alternative environments have delivered limited success. For instance, Microsoft’s Project Natick explored deploying underwater data centers to leverage natural oceanic cooling. However, such subsea deployments

face severe challenges regarding scalability, maintenance, and the environmental impact of thermal pollution [6].

Space-based data centers offer theoretical access to continuous solar energy and passive radiative cooling in the -270°C vacuum of deep space [7, 8]. Industry interest has accelerated along three distinct architectural lines, each making a different bet on how orbital AI should scale.

Monolithic proof-of-concept. Starcloud-1 cleared the most demanding feasibility hurdle to date: an Nvidia H100 GPU trained a nanoGPT model in low Earth orbit, showing that server-class accelerators can withstand launch shock, radiation exposure, and thermal cycling [9]. The company’s longer-term Hyper-cluster concept still concentrates inference and training on a small number of heavy nodes equipped with dedicated cooling and downlink capacity, rather than distributing computation across the full constellation [7]. That design limits inter-satellite coordination, but it also creates a concentrated vulnerability. A single spacecraft failure disables all on-orbit compute, and a single downlink becomes a bottleneck as Earth-observation or inference demand grows.

Modular research clusters. Google Research’s Project Suncatcher pursues a different scaling strategy [10]. Instead of one oversized platform, Suncatcher envisions fleets of smaller satellites carrying radiation-tested Trillium TPUs, flown in close formation. In one reference design, an 81-satellite cluster occupies a 1 km-radius volume so that commercial dense-wavelength-division-multiplexing transceivers can sustain multi-terabit free-space optical ISLs at short range. Prototype launches with Planet Labs are planned by 2027. The program’s immediate research agenda centers on formation-flight control, link-budget engineering, and launch-cost learning curves: the systems questions that must be settled before space-based ML clusters can rival terrestrial pod-level throughput. Federated-learning protocol design under per-node energy budgets lies outside its current scope.

Million-node production constellations. In January 2026, SpaceX filed an application with the Federal Communications Commission for an Orbital Data Center system comprising up to one million LEO satellites [11]. The proposed constellation would operate between 500 and 2,000 km altitude, in both 30° and sun-synchronous inclinations, within narrow 50 km orbital shells. The filing contends that near-constant solar illumination and radiative cooling in vacuum can yield substantial cost and energy savings for large-scale AI inference serving billions of users. Inter-satellite backhaul would draw on SpaceX’s existing Starlink laser-mesh infrastructure. Following the integration of xAI into SpaceX, Elon Musk has argued that orbital compute is the only viable long-term path to scaling AI without further burdening terrestrial power grids. The document is fundamentally an infrastructure and spectrum authorization filing. It specifies orbital geometry, optical links, and deployment scale, but it does not analyze gradient-buffer dynamics, update staleness, or the per-satellite tradeoff between local training and ISL transmission under intermittent power.

These three architectural lines respond to the same terrestrial constraints on energy, water, and land, yet they pose different performance questions. A single heavy node is optimized for hardware survival. A hundred-node TPU cluster is optimized for formation flight and pod-level bandwidth. A million-node constellation is optimized for aggregate inference capacity. None of them characterizes what occurs within each nanosatellite when federated learning must proceed over ISLs that switch intermittently as batteries discharge, eclipses arrive, and optical links fall out of alignment. That per-node scheduling question matters most in the distributed regime that Starlink-scale and Suncatcher-scale designs are meant to enable: thousands of energy-limited clients exchanging compressed gradients instead of raw sensor data.

To realize constellation-scale AI processing, the architecture must move beyond centralized servers and capacity planning toward a federated orbital edge mesh [12]. In this arrangement, a large constellation of nanosatellites performs local machine learning and exchanges only lightweight gradient updates [13] over laser inter-satellite links (ISLs) [14]. DiLoCo-style local steps [15] reduce synchronization frequency by accumulating k gradients before each transmission round, but delay becomes more costly as k grows. A gradient that spends twenty slots in a local buffer may be stale by the time it reaches the aggregator. The central question for this architecture is therefore one of performance engineering: how do energy availability, link intermittency, and queueing dynamics jointly limit training throughput? That question is distinct from, and complementary to, the hardware survival question that Starcloud-1 answered. We develop a stochastic framework for it in the sections that follow.

Scope and methodology. We study a single nanosatellite that trains locally, buffers gradient updates, and transmits them when both battery charge and ISL contact permit service. Time is discretized into slots. The satellite’s battery level, illumination state, and contact state evolve as a finite Markov chain; the

gradient backlog forms a queue whose server is switched on and off by that chain. We derive the effective server availability P_{ready} in closed form, solve the resulting Markov-modulated queue exactly for mean update staleness, extend the analysis to an N -satellite constellation under independent and correlated eclipses, and formulate a constrained Markov decision process (CMDP) for joint train-and-transmit scheduling. All analytical predictions are validated against a discrete-event simulator calibrated to representative CubeSat-class power budgets.

This paper makes four contributions. First, we present a coupled energy-channel Markov model of a single satellite and derive its effective availability P_{ready} in closed form. Second, we provide an exact Markov-modulated solution for the gradient queue whose mean staleness matches simulation across the full stable load range, whereas a single-geometric vacation approximation does not. Third, we analyze constellation-scale aggregation, confirming merged-rate and product-form structure under independent eclipses and identifying the breakdown under correlated illumination. Fourth, we formulate and solve a CMDP bandwidth-allocation policy that outperforms three heuristic baselines on both throughput and energy-floor adherence.

2 Related Work

Our contribution draws on four lines of prior work, supplemented by recent industry initiatives and orbital federated-learning studies reviewed below.

Federated and distributed learning. Federated averaging [13] established the principle of communication-efficient training over decentralized data [16]. Low-communication variants such as DiLoCo [15] extend this idea by performing many local stochastic gradient descent steps before each synchronization round, which is precisely the batch-arrival regime our staleness analysis models. Asynchronous and buffered aggregation schemes, including FedAsync [17] and FedBuff [18], tolerate stale updates rather than waiting for stragglers. In these settings, update staleness (the interval between gradient production and aggregation) is a central performance metric. We formalize staleness as the queue sojourn time of a gradient awaiting ISL transmission. Prior orbital FL work [19, 20] schedules training around deterministic contact windows but does not treat the gradient buffer as a queue served by a stochastic on-off process, nor does it solve for the resulting sojourn-time distribution. That analytical gap motivates the queueing framework developed in Section 4.

Energy-harvesting communication. Early energy-harvesting designs used Harvest-Store-Use (HSU) architectures, where all collected energy is buffered before consumption. Yuan et al. [21] show that HSU bottlenecks throughput because storage conversion is lossy, with practical cells wasting up to 30% of the energy they buffer, and that a Harvest-Use-Store (HUS) policy, which routes harvested power directly to active loads and stores only the net balance, recovers that lost throughput. We adopt the HUS balance so that our orbital energy model reflects realistic storage efficiency.

Queues with vacations and interruptions. A server that is repeatedly switched off is classically analyzed through vacation queues [22, 23], and networks of such queues through Jackson’s product form [24, 25]. When the off-periods are driven by an exogenous Markov chain, as ours are, the exact tool is the matrix-geometric solution for queues in a random environment [26], which we use for the gradient buffer.

Satellite networking and orbital compute. On-satellite computation under tight communication, compute, and energy budgets falls within the broader field of orbital edge computing [12, 27]. Intermittent inter-satellite links are commonly modeled as two-state Gilbert-Elliott channels [28, 29, 30]. Most orbital data-center proposals to date adopt a centralized architecture [7, 9, 31], and terrestrial alternatives such as underwater siting have not demonstrated scalable success [6]. We study a distributed mesh in which each node trains locally and forwards gradients over ISLs, and we quantify how energy availability and link intermittency jointly bound end-to-end training throughput.

Orbital federated-learning protocols. Two recent studies advance the adaptation of federated learning to orbital constraints from complementary directions. Neither provides the queueing-theoretic performance characterization that our work targets.

Jabbarpour et al. [32] conduct the first systematic empirical comparison of centralized and decentralized FL in orbital edge computing. Their simulation uses a 40-satellite Walker-Delta constellation (five orbital planes, eight satellites per plane, altitudes from 600 to 1,800 km) with a ground station in Canberra. Matlab’s Satellite Communications Toolbox supplies deterministic ground-station visit schedules. Four FL variants are evaluated on IID CIFAR-10 with a convolutional network and non-IID EuroSat with ResNet-18: synchronous

centralized FedSync, asynchronous centralized FedAsync, decentralized FedLEO with intra-plane ISLs, and decentralized FELLO with distance-based edge-server clustering. The study sweeps training rounds (20 to 60), local epochs (10 to 30), and batch sizes (8 to 32), recording accuracy, wall-clock training time, and communication power.

Three results bear directly on our model. Communication power exceeds computation power in every tested configuration. FedSync draws 128 kW to reach 80% accuracy on CIFAR-10, while FELLO requires only 17 kW, an 86% reduction obtained by routing aggregation through ISLs rather than the ground station. Ground-station access is sparse: at 600 km altitude, each satellite passes over the ground station only five or six times per day, so 60 FL rounds extend to roughly 20 days on CIFAR-10 and 60 days on EuroSat. Higher altitude increases visit frequency but raises transmit power over longer slant paths. Under non-IID data, cluster size has a larger effect on accuracy than under IID partitioning: FedLEO, with 40 participating satellites, outperforms FELLO, with 18, by more than 3% on EuroSat, whereas the IID gap is below 1%. Jabbarpour’s power model covers only the communication subsystem ($P_{\text{comm}} = P_{\text{base}} + \alpha R + \beta R P_{\text{tx}}$) and leaves onboard GPU draw and battery state dynamics to future work. Staleness appears only as aggregate training latency, not as the sojourn time of an individual gradient behind a stochastically interrupted server.

Kim et al. [33] address the algorithm design side. They propose a modular orbital adaptation framework for retrofitting terrestrial FL algorithms to space-based constraints. The framework imposes three protocol modifications. Client selection must use every available ground contact rather than random subsampling. A training round completes only after all selected clients return their updates, because sparse contact windows preclude partial aggregation within a round. Evaluation clients are chosen independently of training clients each round. The authors apply these rules to FedAvg (yielding FedAvgSat), FedProx (FedProxSat, which permits variable local epoch counts during ground-station outages), and FedBuff (FedBuffSat, which performs asynchronous buffered aggregation with a bounded staleness tolerance).

Two further augmentations exploit predictable orbital mechanics. FLSchedule selects clients with the shortest combined initial-contact and revisit time, reducing idle time by nearly half in the authors’ Walker-Star example. FLIntraCC relays model updates through intra-cluster ISLs when the training satellite lacks a direct ground pass. The experimental campaign spans 768 configurations, varying cluster count (1 to 10), satellites per cluster (1 to 10), and ground-station network size (1 to 13 sites drawn from the International Ground Station network). All adapted variants exceed 80% accuracy on FEMNIST. With scheduling and intra-cluster relay, wall-clock training time for a 100-satellite constellation falls from roughly three months to about ten days, a ninefold reduction. Energy enters the analysis only through CubeSat-class compute assumptions (40 GFLOPS/s) and idle-time fractions. The model does not couple battery charge, illumination, and contact into a single Markov chain, and aggregation remains ground-station-centric throughout.

Table 1 situates these studies alongside the industry architectures discussed in Section I. Jabbarpour’s work identifies which FL variant performs best under which orbital hyperparameters. Kim’s work shows how to modify terrestrial FL protocols for orbital contact patterns. Our work addresses a question neither study resolves: under what load does the per-satellite gradient buffer saturate, and how should a satellite schedule local training against ISL transmission to maximize throughput subject to a hard energy floor? We answer this by solving the Markov-modulated gradient queue exactly, showing that a single-geometric vacation approximation underestimates mean staleness by roughly a factor of two when off-periods are over-dispersed, and optimizing per-satellite scheduling through a constrained MDP that dominates fixed heuristics on the throughput-energy tradeoff.

Table 1: Positioning relative to industry architectures and orbital FL studies. “Server” denotes where model aggregation occurs; staleness denotes how update delay is treated.

Work	Scale	Server	Energy model	Staleness
Starcloud [9]	1 node	Onboard	N/A	N/A
Suncatcher [10]	~100	Ground	Solar/bus	N/A
SpaceX ODC [11]	$\leq 10^6$	Distributed	Solar	N/A
Jabbarpour [32]	40 sats	GS+ISL	Comm. power	Train. time
Kim [33]	≤ 100	Ground	Idle frac.	Buffer
Ours	Per-sat.	ISL mesh	HUS chain	Queue sojourn

3 State Space Formulation

The state of a single satellite at time step t is the joint state vector $S_t = (B_t, I_t, C_t)$, with components as follows.

- $B_t \in \{0, 1, \dots, B_{\max}\}$ is the discrete battery charge level.
- $I_t \in \{0, 1\}$ is the illumination state (0 = eclipsed/discharging, 1 = sunlit/charging).
- $C_t \in \{0, 1\}$ is the ISL contact state (0 = no line-of-sight to next hop, 1 = connected).

The total state space $\mathcal{S}$ has a size of $(B_{\max} + 1) \times 2 \times 2$. Keeping $B_{\max}$ relatively small (e.g., 20 or 50 discrete buckets) will keep the transition matrix tractable for quick numerical evaluation.

To build the full transition matrix P , it is mathematically much cleaner to assume that orbital illumination and ISL topology are independent of the satellite's internal battery. The probability of transitioning from state $s = (b, i, c)$ to state $s' = (b', i', c')$ is:

$$P(S_{t+1} = s' \mid S_t = s) = P(I_{t+1} = i' \mid I_t = i) \cdot P(C_{t+1} = c' \mid C_t = c) \cdot P(B_{t+1} = b' \mid b, i, c)$$

3.1 The Illumination and Contact Sub-Chains

Since physical orbital mechanics and constellation topology are independent of the satellite's internal battery, I_t and C_t are modeled as independent two-state discrete-time Markov chains.

Illumination state (I_t). The satellite alternates between eclipse ($I_t = 0$) and sunlit ($I_t = 1$) phases. The transition matrix P_I is defined using the orbital period parameters:

$$P_I = \begin{bmatrix} 1 - p_{01}^I & p_{01}^I \\ p_{10}^I & 1 - p_{10}^I \end{bmatrix}$$

where p_{01}^I is the probability of exiting the eclipse and p_{10}^I is the probability of entering the eclipse.

Contact state (C_t). Intermittent Laser Inter-Satellite Link (ISL) is modeled using a discrete-time Gilbert-Elliott channel model [29, 30]. The transition matrix P_C is:

$$P_C = \begin{bmatrix} 1 - p_{01}^C & p_{01}^C \\ p_{10}^C & 1 - p_{10}^C \end{bmatrix}$$

where p_{01}^C represents link acquisition and p_{10}^C represents link loss due to optical misalignment or orbital blockages.

3.2 The Battery State Dynamics

The battery transition is deterministic when conditioned on the current state (b, i, c) . To maximize energy efficiency and limit storage conversion losses, we adopt a Harvest-Use-Store (HUS) architecture rather than the traditional Harvest-Store-Use (HSU) model [21]. Under HUS, harvested power is routed directly to active loads during each slot. Only the net surplus or deficit is deposited into or withdrawn from the battery. Yuan et al. show that HSU architectures can waste up to 30% of harvested energy in storage round-trips; the HUS policy recovers that throughput by serving loads before the battery intermediary whenever solar power is available.

Let η denote the solar energy harvesting rate per sunlit slot, P_{base} the baseline avionics load, P_{tx} the ISL optical transmission power, and P_{gpu} the power drawn by the onboard GPU during local training [34]. The net energy change over time step Δt , evaluated before battery routing, is

$$\Delta E(i, c) = \eta \cdot i - P_{\text{tx}} \cdot c - P_{\text{gpu}} \cdot (1 - c) - P_{\text{base}}.$$

The first term captures solar harvesting when $I_t = 1$. The second term charges transmission power when the ISL is active ($C_t = 1$). The third term charges GPU power when the satellite trains locally while the link is

down ($C_t = 0$). The avionics load P_{base} is assessed every slot regardless of activity. When the satellite both trains and transmits in the same slot under the CMDP policy of Section V, both P_{gpu} and P_{tx} contribute.

The battery state at $t+1$ is updated by this net energy, bounded below by zero and above by the physical capacity B_{max} :

$$B_{t+1} = \min(\max(b + \Delta E(i, c), 0), B_{\text{max}}).$$

The conditional transition probability $P(B_{t+1} = b' \mid b, i, c)$ equals one when b' satisfies this update and zero otherwise. The threshold B_{crit} defines a hard energy floor below which the satellite is forbidden to transmit. This reserve protects against deep discharge during extended eclipse sequences and reflects standard battery management practice on power-constrained spacecraft buses.

3.3 Stationary Analysis and Effective Availability

Let P be the full $|\mathcal{S}| \times |\mathcal{S}|$ transition matrix generated by the product of the sub-chains. The stationary distribution, π , is computed by solving the global balance equations:

$$\pi P = \pi \quad \text{subject to} \quad \sum_{s \in \mathcal{S}} \pi_s = 1$$

Here, π_s yields the steady-state probability of the satellite occupying joint state $s = (b, i, c)$. To couple this physical layer model with the gradient aggregation queueing network, the Effective Availability (P_{ready}) is extracted, which is the marginal probability that the satellite is successfully connected and has sufficient charge (above a critical threshold B_{crit}) to transmit its gradient updates:

$$P_{\text{ready}} = \sum_{b=B_{\text{crit}}+1}^{B_{\text{max}}} (\pi_{(b,1,1)} + \pi_{(b,0,1)})$$

Gradients buffer at each satellite and are served only during ISL contact windows. The constellation is modeled as an open network of intermittent-server queues feeding a shared aggregator. To formalize this intermittency, the ISL is characterized using a digital two-state Gilbert-Elliott Markov channel model [28]. Based on the physical phenomena of signal shadowing and multipath fading, the link alternates between a “Good” (aligned/unshadowed) state and a “Bad” (misaligned/shadowed) state. In the unshadowed state, the link supports data transmission with a mean bit error probability typically in the range of 10^{-4} to 10^{-2} , allowing the queue server to process gradient updates at a nominal optical capacity μ . However, when orbital dynamics force a shadowed or misaligned state, the channel bit error probability surges to approximately 0.3. At this threshold, forward error correction fails, and the effective service rate drops to zero.

By mapping the steady-state probabilities P_{ready} derived in Section 3.3 to the server uptime, the stability of the local gradient buffers can be evaluated. The bounds on aggregation delay and update staleness as functions of constellation size, DiLoCo-style local-step count [15], and gradient compression ratio are derived and then used to identify the specific channel regimes where they survive this intermittent service, and the exact fading limits where they break down.

3.4 Relation to Prior Orbital Models

Jabbarpour et al. treat ground-station visibility as a deterministic schedule derived from ephemerides [32]. That representation is well suited to comparing FL protocol variants at the round level, but it smooths over the short-timescale ISL outages that govern gradient-buffer dynamics in a laser mesh. Kim et al. encode contact opportunities as precomputed Systems Tool Kit (STK) access windows fed into a Flower simulator, and they model intra-cluster relay as a scheduling augmentation rather than a stochastic service process [33].

Our Gilbert–Elliott contact chain and coupled battery dynamics retain variability within each ground pass. A satellite may have geometric line of sight to its next hop yet remain unable to transmit because $B_t \leq B_{\text{crit}}$. It may be connected yet unable to serve because the bit-error rate exceeds the forward-error-correction threshold. The resulting off-period has two distinct phases: a fast contact-limited branch, in which the link is down and recovers at rate p_{01}^C , and a slow battery-limited branch, in which the satellite is connected but discharged and must recharge above B_{crit} on the illumination cycle. That two-phase structure over-disperses the off-period relative to a geometric surrogate and motivates the exact Markov-modulated queue analysis in Section 4.

3.5 Calibration

The nominal configuration used throughout our experiments sets $\eta = 4.0$ energy units per sunlit slot, $B_{\max} = 20$, $B_{\text{crit}} = 4$, $P_{\text{base}} = 1.0$, $P_{\text{gpu}} = 2.5$, and $P_{\text{tx}} = 3.0$. These values represent a CubeSat-class bus with a modest onboard GPU and an optical ISL transmitter whose draw exceeds the avionics baseline. Illumination and contact transition probabilities are chosen so that the satellite spends roughly 40% of slots in the ready set $\mathcal{R}$, yielding $P_{\text{ready}} = 0.437$ at the calibrated point. Gradient arrivals occur at rate $\lambda = 0.5$ per slot and the ISL serves at rate $\mu = 2$ per ready slot. Energy, time, and gradient counts are expressed in normalized units; absolute rates are meaningful relative to the model, while cross-configuration trends and policy rankings are the intended outputs.

Table 2: Notation.

Symbol	Meaning
$B_t, B_{\max}, B_{\text{crit}}$	battery level, capacity, energy floor
I_t, C_t	illumination and ISL contact indicators
$Q_t, Q_{\max}$	gradient backlog and buffer cap
η	solar harvesting rate
$P_{\text{base}}, P_{\text{gpu}}, P_{\text{tx}}$	avionics, GPU, and transmit power
P_{ready}	effective server availability
λ, μ	gradient arrival and service rate (per slot)
β	link-recovery rate
ρ_{eff}	effective load $\lambda/(\mu P_{\text{ready}})$
$\mu_{\text{eff}}, \mu_{\text{agg}}$	compression-boosted and aggregator service rates
k	DiLoCo local-step (batch) size
N	constellation size
θ, γ	Lagrange multiplier and MDP discount factor

4 Gradient Queue Analysis

4.1 Single-Satellite Gradient Queue

Gradients arrive at satellite i 's local buffer as a Poisson process of rate λ gradients per slot, matching the arrival model in the simulator. The buffer is served at rate μ gradients per slot whenever the satellite is in the ready set $\mathcal{R} = \{(b, i, c) : b > B_{\text{crit}}, c = 1\}$ and is not served otherwise. The server uptime is the effective availability P_{ready} derived in Section 3.3. The backlog therefore evolves as a discrete-time queue whose server is switched on and off by the energy-channel chain,

$$Q_{t+1} = \min(\max(Q_t + A_t - \mu \mathbf{1}[\text{ready}_t], 0), Q_{\max}),$$

with $A_t \sim \text{Poisson}(\lambda)$. This is the canonical time-slotted queue with a random environment.

Because the battery and contact states evolve jointly, the off-period (a sojourn outside $\mathcal{R}$) is not a simple geometric. We first summarize it by a single recovery rate β , the stationary-weighted probability that a not-ready state steps into $\mathcal{R}$ in one slot,

$$\beta = \frac{\sum_{s \notin \mathcal{R}} \sum_{s' \in \mathcal{R}} \pi_s P_{ss'}}{\sum_{s \notin \mathcal{R}} \pi_s},$$

which gives a mean off-period $1/\beta$. As we show below, the true off-period is in fact two-phase, with a fast contact-limited branch and a slow battery-limited branch, and its dispersion is what governs staleness.

4.2 Stability and Mean Staleness

The queue is stable if and only if the effective load satisfies

$$\rho_{\text{eff}} = \frac{\lambda}{\mu \cdot P_{\text{ready}}} < 1,$$

since the energy-channel chain is finite and ergodic with long-run service capacity μP_{ready} , and a mean-drift argument gives positive recurrence exactly when $\lambda < \mu P_{\text{ready}}$.

A first closed-form estimate treats the off-period as geometric with rate β and applies the M/G/1 multiple-vacation decomposition [22, 23], giving

$$\mathbb{E}[T] \approx \frac{1}{\mu - \lambda} + \frac{2 - \beta}{2\beta}, \quad (1)$$

where the first term is an idealized always-on M/M/1 sojourn and the second is the residual off-period a gradient waits for the link to recover. Mean queue length follows from Little’s law, $\mathbb{E}[L] = \lambda \mathbb{E}[T]$ [25]. We treat Equation (1) only as a continuous-time reference approximation, since the multiple-vacation discipline assumes vacations begin at queue emptying whereas our server is interrupted mid-busy-period by an exogenous chain, the geometric assumption discards the off-period variance, and the discrete-time server drains up to μ gradients per ready slot rather than one customer at a time.

That variance is large. At the calibrated configuration ($\eta = 4.0$, $B_{\text{max}} = 20$, $B_{\text{crit}} = 4$), the off-period distribution of the energy-channel chain has mean $\mathbb{E}[V] = 20.5$ slots and squared coefficient of variation $\text{cv}^2 = 1.98$. A geometric off-period of the same mean would have $\text{cv}^2 = 1 - \beta = 0.95$. The excess dispersion arises from a genuine two-phase structure. Of the stationary not-ready mass, 59% lies in contact-limited states where the link is down and recovers at rate p_{01}^C . The remaining 41% lies in battery-limited states where the satellite maintains contact but cannot transmit because $B_t \leq B_{\text{crit}}$, and must wait for a sunlit interval to recharge. Collapsing these two recovery timescales into a single rate β therefore understates the tail of the waiting-time distribution and, by extension, the mean queue sojourn.

To obtain the exact answer we solve the joint (Q, B, I, C) chain directly. The process is a quasi-birth-death chain in the backlog level with the energy-channel state serving as the phase: a queue in a random environment of the matrix-geometric type [26]. We compute the stationary distribution by solving the global balance equations with a sparse linear solver, truncating the backlog at a level where the boundary mass falls below 10^{-4} . The exact mean backlog follows directly, and Little’s law yields the exact mean staleness.

Table 3 and Figure 1 report the comparison against simulation. At the nominal load $\rho_{\text{eff}} = 0.572$, the exact solve gives $\mathbb{E}[T] = 38.9$ slots, matching a 16-seed Poisson-driven rollout at 39.1 ± 2.9 slots. The geometric estimate of Equation (1) returns only 20.7 slots, nearly a factor of two below both the exact value and the simulation. Across the stable range $\rho_{\text{eff}} \in [0.1, 0.8]$, the exact solve tracks simulation within the confidence interval at every tested load, while the geometric estimate remains near 20 slots regardless of load. It therefore underestimates staleness at low load and overestimates the rate at which staleness grows at high load.

The practical implication is that mission planners who size gradient buffers or set DiLoCo local-step counts using a single recovery rate β will underestimate the buffer occupancy needed to absorb ISL outages. An earlier validation pass drove arrivals through the contact-gated policy, under which gradients are generated only while the link is down; that configuration realized an effective arrival rate of approximately 0.10 gradients per slot rather than λ . The rollout reported here injects an independent Poisson(λ) stream so that the analytical model and the simulator are matched on load.

4.3 N-Satellite Constellation

For N satellites with independent energy-channel chains and i.i.d. Poisson(λ) arrivals, the aggregator at the parameter server receives the superposition of N on-off gradient streams. Two structural properties follow. First, the merged arrival rate equals $N\lambda$ by additivity of the intensities of N independent Poisson streams; in steady state each per-satellite server is work-conserving, so its long-run departure rate also equals λ and the superposed departure intensity is $N\lambda$. Second, because the per-satellite chains, arrivals, and buffers are mutually independent and do not interact, the joint backlog factors as $\pi(q_1, \dots, q_N) = \prod_i \pi_i(q_i)$. This

factorization is the trivial consequence of independence rather than Jackson’s product-form theorem [24], which couples queues through routing and relies on Poisson departures.

The aggregator is nonetheless not exactly M/M/1, because the departures of an on-off server are bursty rather than Poisson. Service is suspended during off-periods and clusters during on-periods, so Burke’s theorem does not apply. By the Palm-Khinchine superposition theorem the merged stream approaches Poisson as N grows, provided each contributing stream is asymptotically sparse and none dominates, and under that condition an M/M/1 aggregator model becomes reasonable. Section 6 confirms these points empirically and shows that even at independence the aggregator delay exceeds the naive $1/(\mu_{\text{agg}} - N\lambda)$ prediction, a direct symptom of departure burstiness.

4.4 Breakdown Regimes

The independence behind the product form fails in two regimes that the model makes precise.

Correlated eclipses. Satellites in the same orbital plane experience eclipse entry and exit at nearly the same orbital phase, so their illumination chains are positively correlated in practice. We introduce a correlation parameter $\rho_{\text{corr}} \in [0, 1]$, defined as the per-slot probability that all satellites in a plane adopt the same illumination transition simultaneously. When $\rho_{\text{corr}} = 0$, illumination evolves independently and the product-form factorization of Section 4 holds exactly. As ρ_{corr} increases, per-satellite queue lengths become positively correlated and aggregation delay at the parameter server exceeds the independent baseline. At $\rho_{\text{corr}} = 1$, all satellites in the plane share a single illumination process. Section 6 reports that aggregation delay rises to roughly 3.6 times the independent value at full correlation, which is the operational consequence of assuming independence in co-planar constellations such as those studied by Kim et al. [33] and Jabbarpour et al. [32].

DiLoCo-style batch transmission [15]. Under k local stochastic gradient descent (SGD) steps before a sync round, gradients arrive in batches of size k , giving a batch-arrival M^[X]/M/1 queue with batch rate λ/k [35]. Applying Little’s law to the standard batch result $\mathbb{E}[L] = \rho/(1 - \rho) + \Lambda k(k - 1)/(2\mu(1 - \rho))$, with $\Lambda = \lambda/k$ and $\rho = \lambda/\mu$, the staleness gains an additive term

$$\mathbb{E}[T]_{\text{batch}} = \mathbb{E}[T] + \frac{k - 1}{2(\mu - \lambda)}.$$

The correction grows linearly in k and diverges as $\lambda \rightarrow \mu$, imposing a firm ceiling on how many local steps can be exchanged for fewer communication rounds before staleness overwhelms the training benefit. The off-period contribution is unchanged under batching, because batch composition does not alter the link and energy dynamics.

5 Bandwidth-Allocation Policy Optimization

The queueing analysis in Section 4 characterizes gradient staleness under fixed scheduling rules. A satellite that always trains when the link is down and always transmits when it is up may deplete its battery during a long eclipse. A satellite that gates all training on link unavailability conserves power but underutilizes sunlit slots when the buffer is nonempty. These tensions suggest that per-slot scheduling deserves explicit optimization rather than a fixed heuristic.

5.1 CMDP Formulation

We formulate the per-satellite scheduling problem as a constrained Markov decision process (CMDP) [36]. The state at time t is $S_t = (B_t, I_t, C_t, Q_t)$, where $Q_t \in \{0, \dots, Q_{\text{max}}\}$ is the local gradient buffer occupancy. At each slot the satellite selects one of four action pairs $(a_{\text{local}}, a_{\text{tx}}) \in \{0, 1\}^2$: idle (0, 0), train-only (1, 0), transmit-only (0, 1), or train-and-transmit (1, 1). Training draws power P_{gpu} whenever $a_{\text{local}} = 1$ and the link is down; transmission draws P_{tx} whenever $a_{\text{tx}} = 1$ and the link is up.

The immediate reward $R(s, a)$ equals the number of gradients drained from the buffer in slot t . The immediate cost $\mathcal{C}(s) = \mathbf{1}[B_t < B_{\text{crit}}]$ is an energy-floor indicator that depends on the state but not on the action: crossing below B_{crit} incurs the same penalty regardless of what the satellite attempted. The CMDP

seeks a policy π that maximizes long-run throughput while keeping the time-averaged cost below a tolerance ε :

$$\max_{\pi} \mathbb{E}_{\pi}[R] \quad \text{subject to} \quad \mathbb{E}_{\pi}[C] \leq \varepsilon.$$

5.2 Heuristic Baselines

Three fixed policies anchor the comparison. The **power_oblivious** policy trains and transmits whenever the ISL is available, mirroring the synchronous participation pattern of FedSync in Jabbarpour et al.’s study [32]. The **contact_gated** policy trains only while the link is down and transmits only while it is up, enforcing a strict alternation that conserves energy at the cost of idle sunlit slots. The **power_weighted** policy scales training and transmission probabilities by the battery level relative to $B_{\max}$, approximating the HUS balance of Section 3.3 without solving for the optimal action in each state. None of these rules conditions on buffer occupancy Q_t , which the CMDP exploits to drain backlogs aggressively when power headroom permits.

5.3 Lagrangian Relaxation and Value Iteration

We relax the energy-floor constraint with a Lagrange multiplier $\theta \geq 0$ (distinct from the arrival rate λ in Section 4), obtaining an unconstrained discounted MDP:

$$V^{\theta}(s) = \max_a \left[R(s, a) - \theta C(s) + \gamma \sum_{s'} P(s' | s, a) V^{\theta}(s') \right].$$

Value iteration with discount factor $\gamma = 0.99$ [37] solves this Bellman equation. Sweeping θ from 0 (pure throughput) to 25 (strong energy conservation) traces the tradeoff curve between mean throughput and energy-floor violation frequency. The joint (B, I, C, Q) state space contains $(B_{\max} + 1) \times 2 \times 2 \times (Q_{\max} + 1) = 21 \times 4 \times 21 = 1,764$ states with queue cap $Q_{\max}^{\text{CMDP}} = 20$. Each value-iteration pass is a vectorized update over this space and converges in fewer than 1,000 iterations in our implementation.

6 Results

6.1 Simulation Environment

All simulation results reported below are produced by the orbital federated-learning simulator described in our public repository. The simulator advances the energy-channel Markov chain and gradient buffer in discrete time steps. At each slot it samples illumination and contact transitions from P_I and P_C , updates the battery under the HUS rule of Section 3.3, injects Poisson(λ) gradient arrivals, and drains up to μ gradients when the satellite occupies the ready set $\mathcal{R}$. For constellation experiments, N independent satellite chains run in parallel and their departure streams are merged at a single aggregator whose service rate is μ_{agg} . Eclipse correlation is controlled by the parameter ρ_{corr} , which governs the probability that satellites in the same plane adopt identical illumination transitions in a given slot. Policy rollouts for the CMDP comparison execute the learned action map or a fixed heuristic over 5,000 slots per seed. Each experiment is repeated over 20 independent random seeds, and we report means with 95% confidence intervals computed across seeds. Analytical quantities (P_{ready} , β , $\mathbb{E}[T]$) are computed directly from the transition matrix and the joint queue balance equations without sampling.

6.2 Queueing Model Validation

Table 3 reports results for the calibrated configuration ($\eta = 4.0$, $B_{\max} = 20$, $B_{\text{crit}} = 4$, $\lambda = 0.5$, $\mu = 2$). The geometric off-period estimate from Equation (1) returns 20.7 slots. Because the true off-period is over-dispersed ($\text{cv}^2 \approx 2.0$), this approximation is unreliable. The exact Markov-modulated solution gives $\mathbb{E}[T] = 38.9$ slots and agrees with a 16-seed Poisson-driven simulation at 39.1 ± 2.9 slots.

Placed in the context of orbital FL, a mean staleness of 39 slots at $\lambda = 0.5$ gradients per slot means that each synchronization-bound update waits nearly twice as long as a naive vacation model predicts. Kim et al.’s buffered aggregation [33] and Jabbarpour et al.’s asynchronous centralized methods [32] are designed

for precisely this regime of delayed updates, yet neither study reports the buffer occupancy distribution that motivates those design choices. Our exact solve supplies that quantity.

Table 3: Stationary and queueing results for the calibrated configuration ($\eta = 4.0$, $B_{\max} = 20$, $B_{\text{crit}} = 4$, $\lambda = 0.5$, $\mu = 2$). The exact Markov-modulated solve matches simulation; the single-geometric vacation estimate does not.

Quantity	Value
P_{ready}	0.437
β (recovery rate)	0.0488
ρ_{eff}	0.572
Off-period $\mathbb{E}[V]$, cv^2	20.5 slots, 1.98
$\mathbb{E}[T]$ vacation estimate (geometric β)	20.7 slots
$\mathbb{E}[T]$ exact Markov-modulated solve	38.9 slots
$\mathbb{E}[T]$ simulation (16 seeds)	39.1 ± 2.9 slots

Figure 1 shows the gradient queue model. Panel (b) is the central validation. Across the stable range $\rho_{\text{eff}} \in [0.1, 0.8]$ the exact solve tracks simulation within the confidence interval at every load, with mean staleness rising from 19 to 83 slots, while the geometric vacation estimate stays near 20 slots and is wrong in both directions. Panel (c) shows the backlog tail $P[Q \geq x]$ decaying geometrically, and panel (d) the DiLoCo batch penalty. The batch penalty grows linearly in the local-step count k , establishing a firm limit on how aggressively a constellation can adopt DiLoCo-style communication reduction before staleness degrades convergence. Orbital FL schedulers [33, 32] implicitly accept this tradeoff when they increase local epoch counts, but they do not bound the penalty analytically.

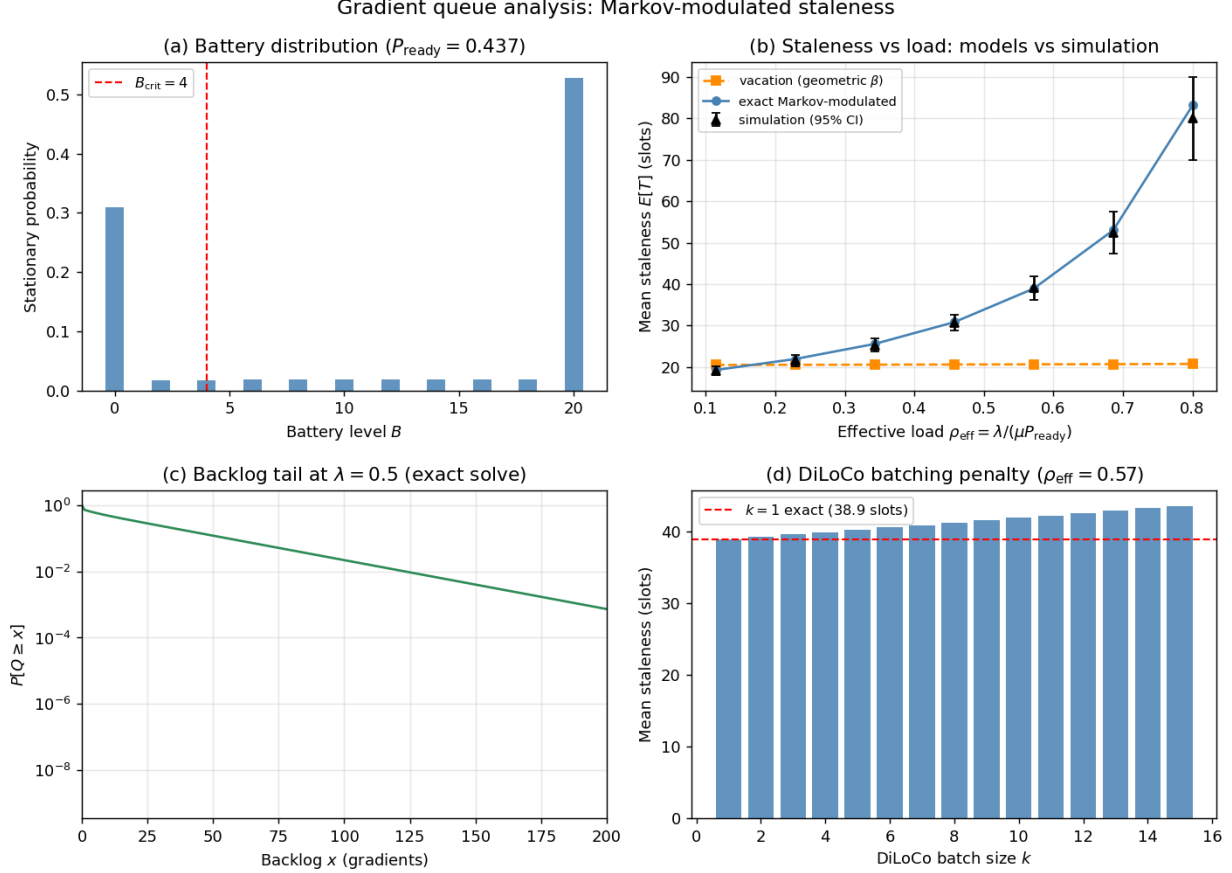


Figure 1: Gradient queue analysis. (a) Stationary battery distribution with P_{ready} marked. (b) Mean staleness vs. effective load. The exact Markov-modulated solve matches simulation (95% CI), while the single-geometric vacation estimate stays flat near 20 slots. (c) Backlog tail $P[Q \geq x]$ at $\lambda = 0.5$. (d) DiLoCo batching penalty vs. local-step count k .

6.3 Constellation Product Form and Breakdown

Figure 2 examines aggregation at the parameter server when N satellites contribute gradient updates.

Panel (a) confirms flow balance: the merged arrival rate at the aggregator equals $N\lambda$, as predicted by the superposition of N independent Poisson streams. A naive M/M/1 model at that arrival rate, however, underestimates aggregation delay. Even when satellites evolve independently, the measured delay exceeds the prediction $1/(\mu_{\text{agg}} - N\lambda)$ because on-off departure processes are bursty rather than Poisson. Service clusters during contact episodes and ceases entirely during off-periods, so Burke’s theorem does not apply and the aggregator sees correlated bursts of traffic.

Panels (b) through (d) trace the effect of eclipse correlation. When $\rho_{\text{corr}} = 0$, per-satellite queues are independent and the joint backlog distribution factorizes as a product of marginals. As ρ_{corr} rises, queue lengths across satellites become positively correlated. At $\rho_{\text{corr}} = 1$, the per-satellite queue correlation reaches 0.70, aggregation delay climbs to roughly 3.6 times the independent baseline, and the total-variation distance between the true joint distribution and the product of marginals grows in tandem. This breakdown is not a modeling artifact. Satellites in the same orbital plane share eclipse transitions by construction, so the independence assumption used in both Kim et al.’s scalability analysis [33] and Jabbarpour et al.’s per-orbit clustering [32] can understate aggregation delay by more than a factor of three in co-planar Walker-Star and Walker-Delta geometries. Constellation designers who size aggregator capacity from independent-satellite

models should apply a correlation margin when satellites share an orbital plane.

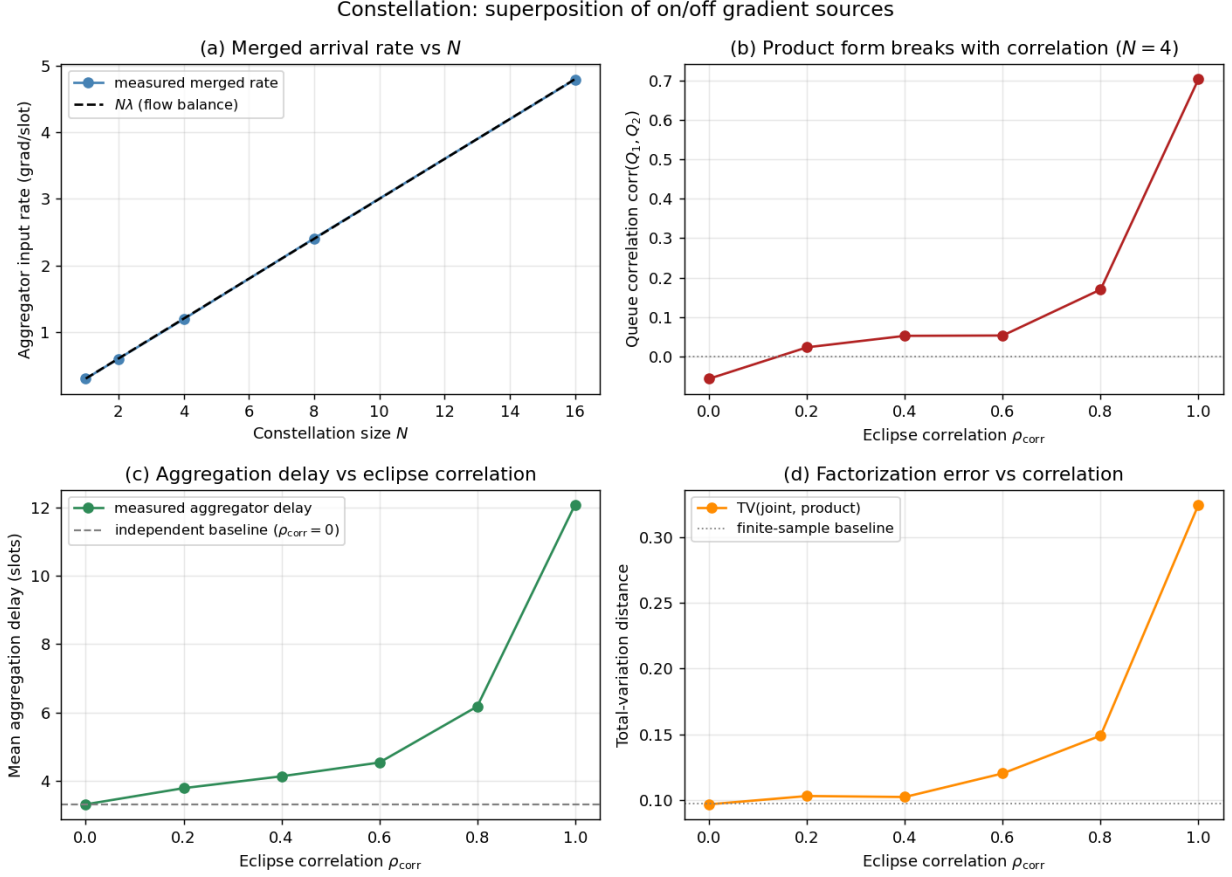


Figure 2: Constellation analysis. (a) Merged arrival rate equals $N\lambda$ (flow balance). (b) Per-satellite queue correlation vs. eclipse correlation ρ_{corr} ($N = 4$). (c) Aggregation delay rises above the independent baseline as eclipses correlate. (d) Total-variation distance from the product of marginals.

6.4 Robustness Under Compression and Fading

Figure 3 sweeps two design parameters that mission designers can influence at the protocol layer.

Gradient compression increases the effective service rate μ_{eff} by reducing the number of bits per update. Panel (a) shows that staleness declines with compression but with sharply diminishing returns, approaching a floor of approximately 17.6 slots. Below that floor, faster service cannot help because the dominant delay is the wait for the energy-channel chain to re-enter the ready set, not the time to drain the buffer once contact resumes. Designers who invest in higher-rate ISLs or more aggressive gradient quantization therefore encounter a point of diminishing returns set by energy and link availability rather than by raw channel capacity.

Panel (b) tracks P_{ready} as the ISL link-loss probability increases. Each increment in loss probability shortens contact episodes and lowers the steady-state fraction of slots in which the satellite can transmit. Panel (c) shows the corresponding effect on mean staleness. At a link-loss probability of approximately 0.16, the effective load ρ_{eff} reaches unity and the queue becomes unstable. This fading limit defines the operating envelope within which the queueing model and the CMDP policy remain valid. Infrastructure filings for large-scale orbital compute [11] and formation-flight studies for TPU clusters [10] specify link capacity targets but do not map those targets to the stability boundary derived here.

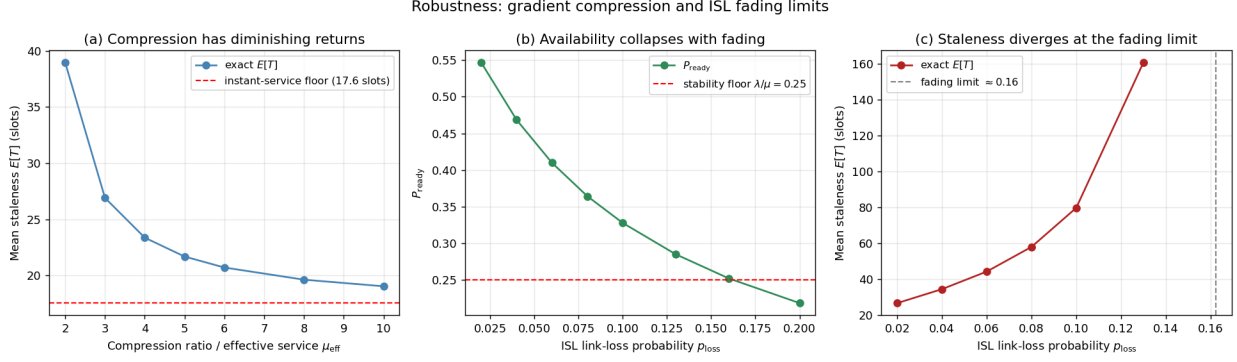


Figure 3: Robustness sweeps. (a) Staleness vs. gradient compression, with the instant-service floor. (b) P_{ready} vs. ISL link-loss probability, with the stability floor λ/μ . (c) Staleness diverges at the fading limit.

6.5 CMDP Policy Comparison

Table 4 compares the three heuristic baselines against the CMDP policy over 20 random seeds with 95% confidence intervals. Each rollout spans 5,000 slots under the calibrated energy model of Section 3.3.

The **power_oblivious** policy achieves the highest raw activity but violates the energy floor in nearly half of all slots (cost fraction 0.479). The **contact_gated** policy is the most conservative, draining only 0.109 gradients per slot because it refuses to train during contact windows even when the buffer is full and the battery is charged. The **power_weighted** heuristic offers a better balance at 0.284 gradients per slot with a 3.7% floor-violation rate, but its worst seed still crosses below B_{crit} in 8.1% of slots.

For any $\theta \geq 0.05$, the CMDP policy incurs zero energy-floor violations across every seed while sustaining 0.326 gradients per slot, a 14.7% throughput improvement over the best heuristic. At $\theta = 0$, the CMDP maximizes throughput at 0.328 gradients per slot but accepts a 27.5% floor-violation fraction. The operating point at $\theta = 0.05$ is the most relevant for a deployed bus with a hard battery reserve requirement: it matches the throughput of the unconstrained optimum while eliminating floor crossings entirely. This gain requires explicit optimization over the joint (B, I, C, Q) state. Fixed scheduling rules, including the orbital FL protocols of Kim et al. [33], do not condition on buffer length and therefore cannot recover the same headroom.

Table 4: Policy throughput and energy-floor fraction (5000-step rollout, $\eta = 4.0$, 20 seeds, mean $\pm$ 95% CI).

Policy	Throughput (grad/slot)	Cost fraction
power_oblivious	0.246 ± 0.011	0.479 ± 0.020
contact_gated	0.109 ± 0.005	0.317 ± 0.019
power_weighted	0.284 ± 0.008	0.037 ± 0.010
CMDP ($\theta = 0$)	0.328 ± 0.007	0.275 ± 0.016
CMDP ($\theta = 0.05$)	0.326 ± 0.010	0.000 ± 0.000

Figure 4 plots the throughput-energy tradeoff curve with confidence intervals. The curve collapses to two CMDP operating points because the learned policy maintains $B > B_{\text{crit}}$ whenever any cost penalty is active. Both CMDP points lie above and to the left of every heuristic, meaning that structured optimization achieves equal or higher throughput at equal or lower energy-floor cost. No heuristic point is undominated by a CMDP point.

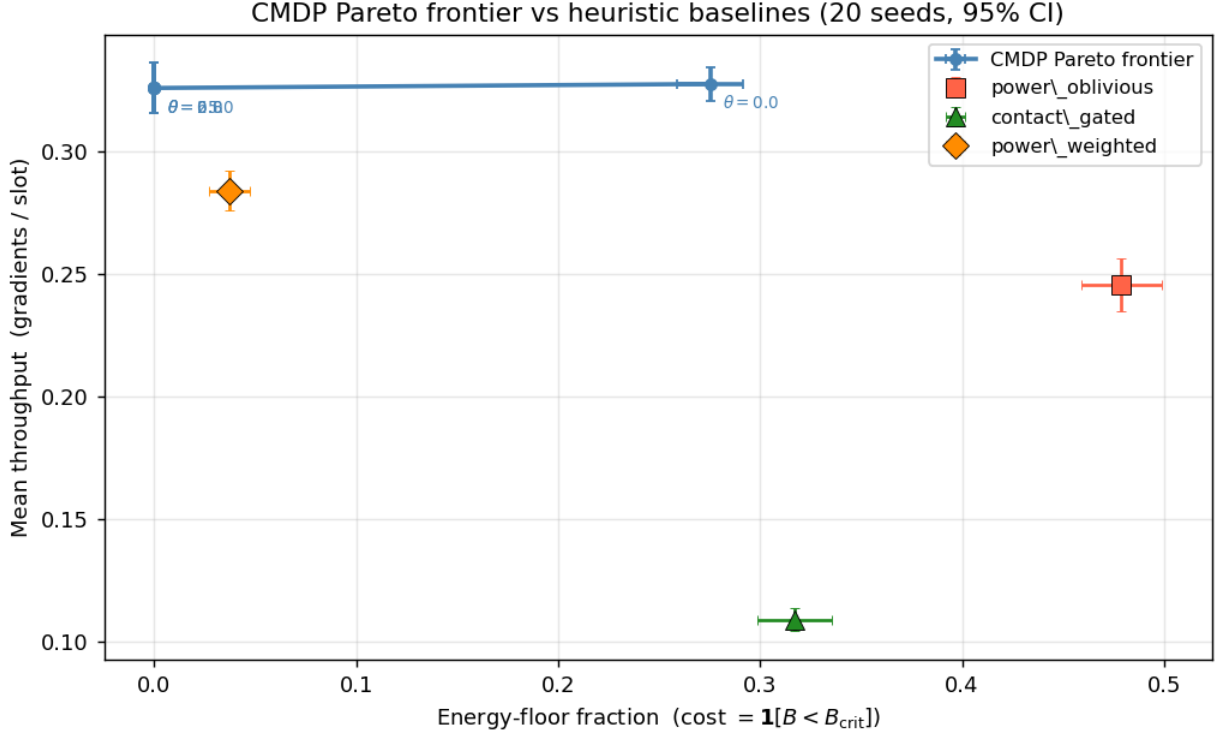


Figure 4: CMDP Pareto frontier (throughput vs. energy-floor fraction) over 20 seeds with 95% confidence intervals, against the three heuristic operating points. Both CMDP points dominate all baselines.

7 Conclusion

We have presented an analytical and computational framework for energy-aware gradient scheduling aboard an orbital nanosatellite engaged in federated learning. The motivation is a gap in the current orbital-AI literature. Industry filings and long-horizon research programs now specify how many satellites to deploy and what hardware to place on orbit [11, 10, 9]. Recent orbital FL studies specify which training protocols to adopt [32, 33]. Neither line of work yet characterizes when a per-satellite gradient buffer saturates under coupled energy, illumination, and link dynamics, nor how a satellite should schedule local training against ISL transmission to maximize throughput subject to a battery reserve.

Our framework addresses that gap in four stages. We model the satellite as a finite Markov chain over battery, illumination, and contact states, and derive the effective server availability P_{ready} in closed form. We solve the resulting Markov-modulated gradient queue exactly and show that mean update staleness matches discrete-event simulation across the full stable load range. A single-geometric vacation approximation underestimates staleness by roughly a factor of two because the off-period is over-dispersed across contact-limited and battery-limited phases. At the constellation level, we confirm that merged arrival rates and product-form backlog structure hold under independent eclipses and fail when satellites in the same orbital plane share illumination transitions. Robustness sweeps identify a compression floor below which faster ISL service yields no further staleness reduction, and a fading limit beyond which the queue becomes unstable. Finally, a CMDP solved by Lagrangian relaxation and value iteration delivers 14.7% higher throughput than the best fixed heuristic at zero energy-floor violations across all simulation seeds.

Per-satellite scheduling optimization is complementary to constellation-scale capacity planning. SpaceX’s million-satellite filing and Google’s Suncatcher cluster design address aggregate compute and link capacity. Jabbarpour et al. and Kim et al. address protocol selection and orbital adaptation. Our results show that even with a fixed protocol and a fixed hardware budget, substantial throughput gains remain available

through state-dependent scheduling over the joint battery, link, and buffer state.

Limitations. The model is deliberately abstract. Energy, time, and gradient counts are expressed in calibrated units rather than fit to a specific satellite bus or orbit. Absolute rates such as gradients per slot are therefore meaningful only in relative terms; the qualitative trends and the ranking of policies are the primary outputs. The exact queue solution truncates the backlog at a finite level chosen so that boundary mass is negligible, rather than computing a closed-form matrix-geometric rate. The CMDP optimizes a single satellite and trades throughput against an energy floor. It does not model downlink contention, coordinated multi-satellite action, or competing objectives such as model accuracy. Gradient compression enters only through an effective service rate, without an accuracy penalty. The constellation study varies eclipse correlation while holding per-link contact processes independent. Radiation-induced bit flips, thermal throttling of the onboard accelerator, and orbit-specific ephemeris data are also outside the current scope.

Future work includes a matrix-geometric treatment of the joint queue that removes backlog truncation, extension of the CMDP to coordinated multi-satellite policies under correlated eclipse schedules, and online adaptation of the Lagrange multiplier as orbital parameters drift over a mission lifetime.

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